Analysis of stress in residual limb of upper leg amputee with prosthetic

ME6124 - Finite Element Method

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INTRODUCTION AND BACKGROUND

- Inspiration drawn from interest in biomedical devices and the application of FEM to prosthetic limbs

- Desire to investigate different type of materials
  - Investigating hyperelastic materials: fits for the soft tissues of the limb
  - Investigating different type of hard materials: fits for the socket component of the prosthetics

- Opportunity to investigate contacts and how parts interact with one another

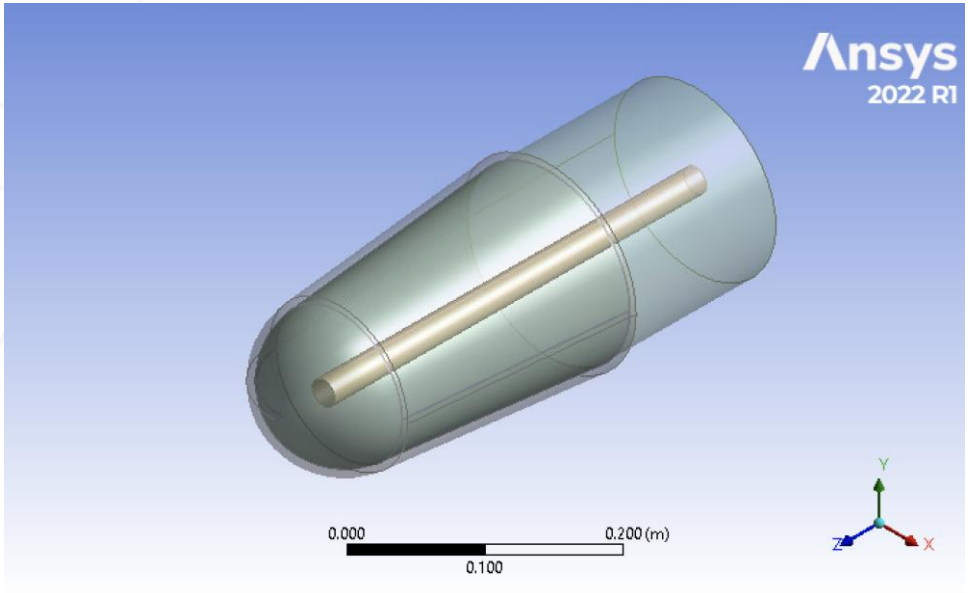
- Motivation to develop a better understanding of the biomechanics of prosthetic limb design and improve patient outcomes

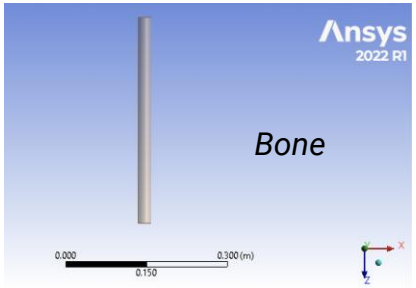
OBJECTIVES

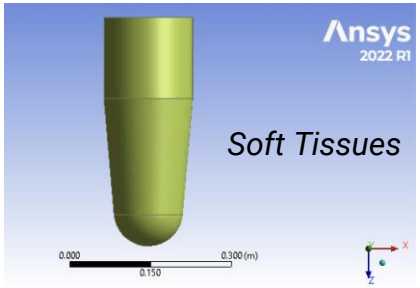
- Create our 3D model consisting in four parts: the bone, the soft tissue, the sock and the socket
- Investigate and apply the loading conditions (weight of the amputee)
- Analyze material properties and model the soft tissue as an isotropic linear elastic material as a first step
- Run simulation and conduct mesh convergence
- Perform further testing and what ifs analysis, varying the materials utilized

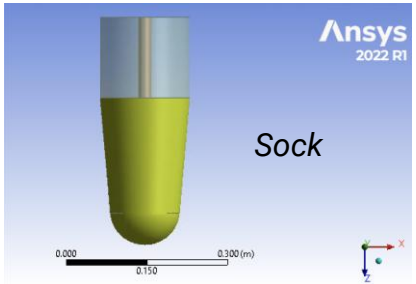
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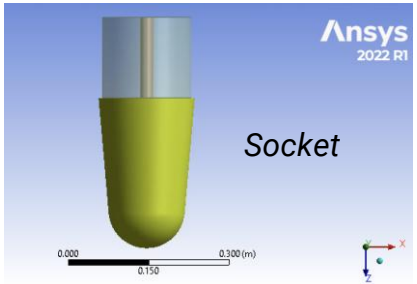
3D MODELING







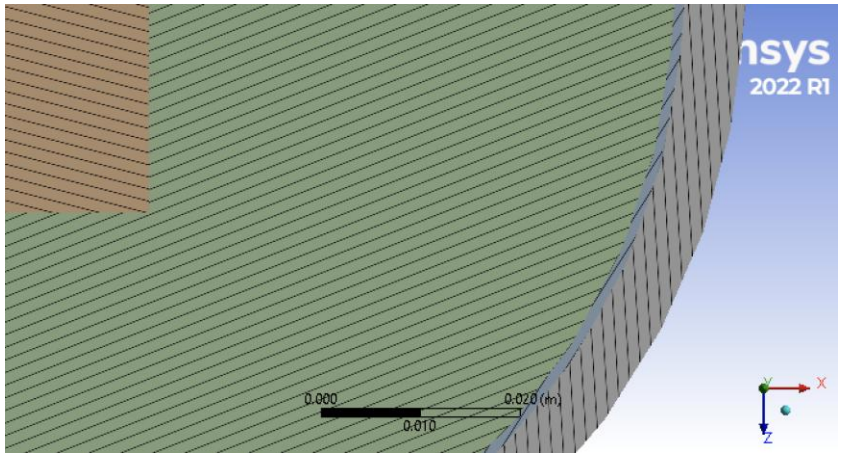




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DESCRIPTION OF EACH PART

- **Bone**: modeled as a cylinder
- **Soft tissues**: include the fat, the skin and the muscles. Modeled as a symmetric rounded cone.
- **Sock**: added to simulate an extra material layer used for increased patient comfort. Modeled as a thin layer covering the residual limb
- **Socket**: modeled as a rigid hollow rounded cone, designed based on the patient's residual limb shape in order to fit over the sock and residual limb



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LOADING CONDITIONS

- Static study of an average sized 80kg male standing fully on prosthetic leg
- The force was assumed to be uniformly distributed on the bone, which resulted in a pressure acting on the top surface of the bone in our 3D model
- Force acting on the top surface of the bone = 784.8N
- Pressure acting on the top surface of the bone = 1.84MPa

BOUNDARY CONDITIONS AND CONTACT DEFINITION

- Fixed at lower part of socket- simulate stationary position
- Rigid contact
  - Bone to soft tissue
- Frictional contact with coefficient of friction as .2
  - Soft tissue to sock
  - Sock to socket
  - Modified from .4 found from literature
  - Target needs to be the harder material

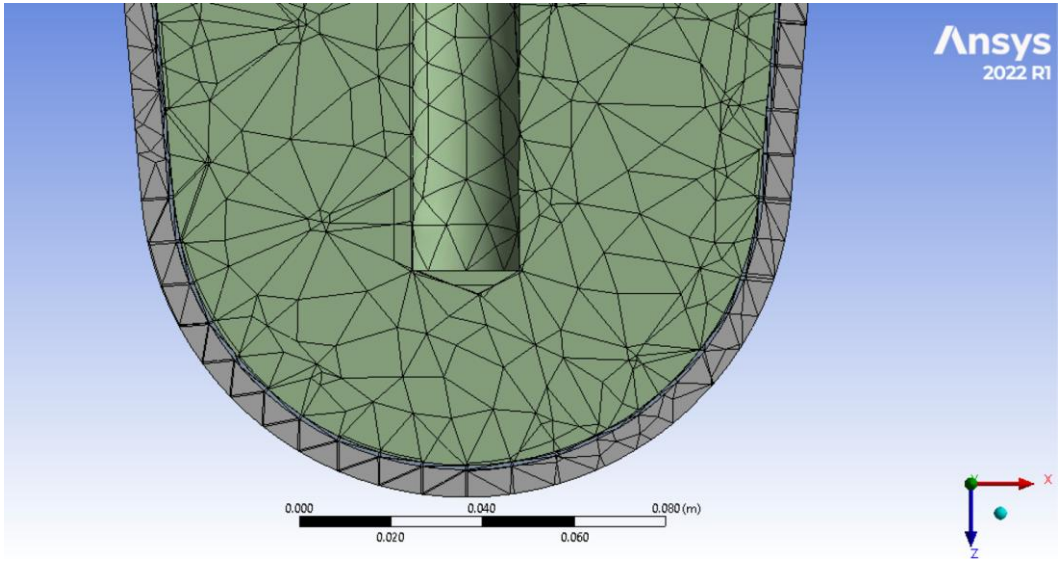
INITIAL SIMULATION - MATERIAL PROPERTIES

- **Bone**: isotropic elastic material with a Young's Modulus of E=15GPa, and a Poisson's ratio of 0.3
- **Soft tissues:** first modeled as an isotropic elastic material. The Young's Modulus was set to E = 1.5MPa, and the Poisson's ratio was set to 0.49
- **Sock:** initially modeled as acrylic lycra material with Young's Modulus .38 MPa and Poisson's ratio set to .49
- **Socket:** modeled as an isotropic elastic material, and more specifically as isotropic elastic propylene. Young's modulus was set to 15 GPa, and the Poisson's ratio was set to 0.3

INITIAL SIMULATION - MESH

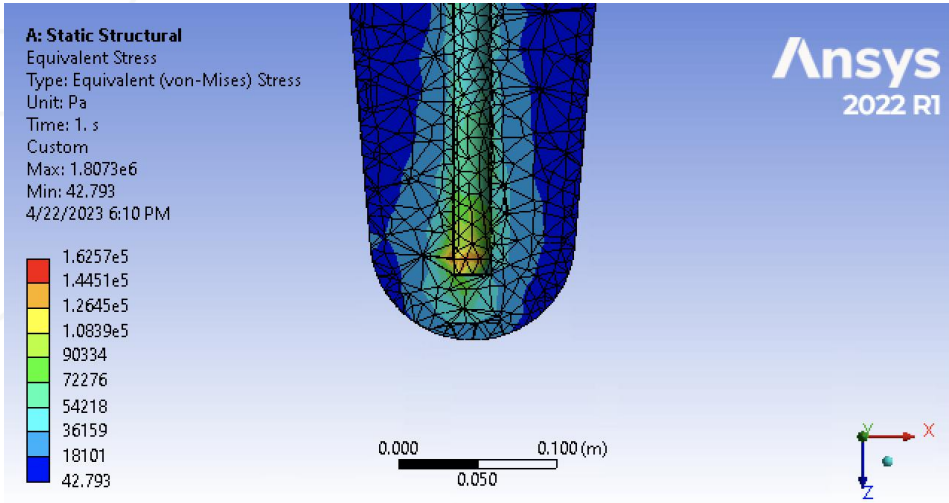
- All triangular elements: although quad elements lead to less errors, the curvature was too complex

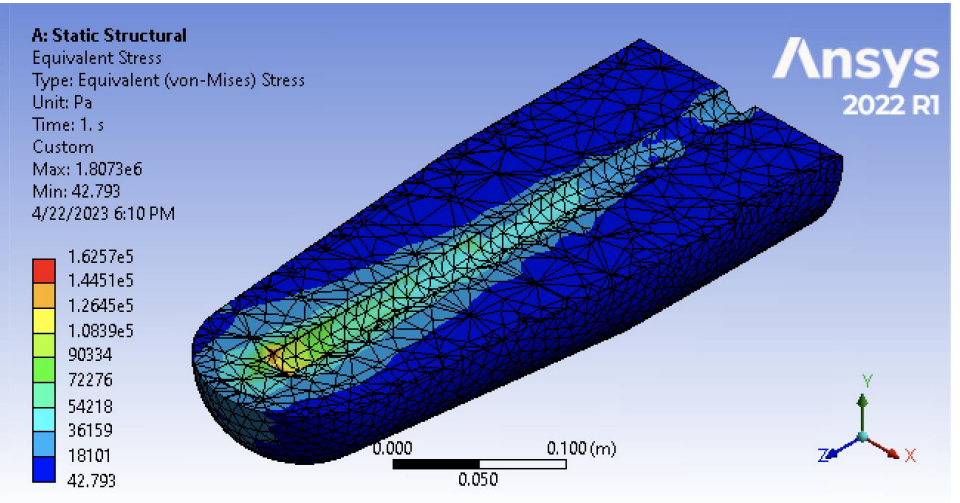
- Coarse mesh for the first simulation



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INITIAL SIMULATION - RESULTS

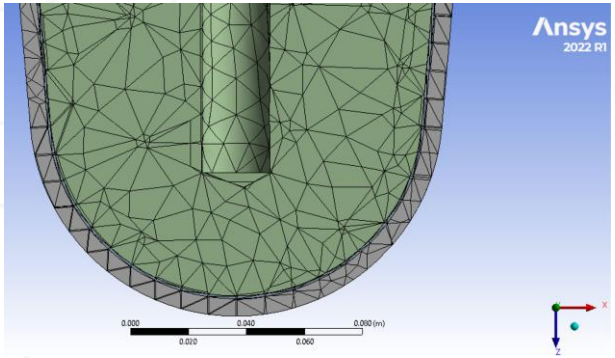




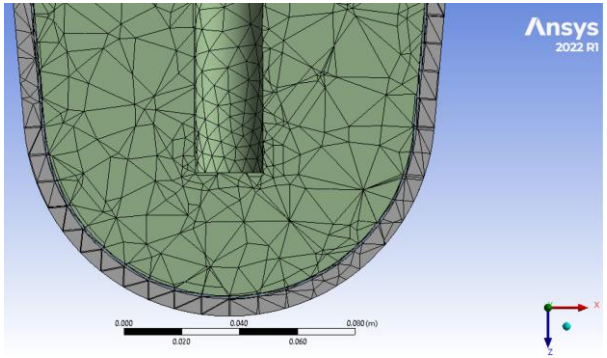
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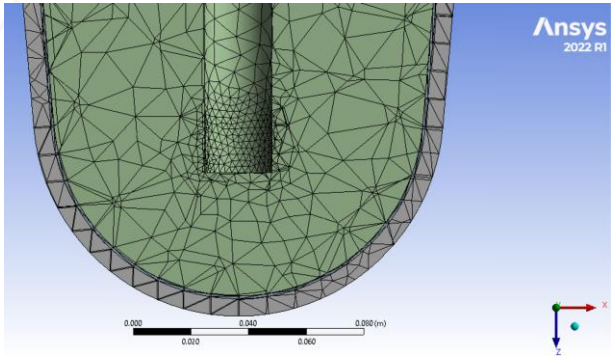
MESH CONVERGENCE STUDY

10 mm

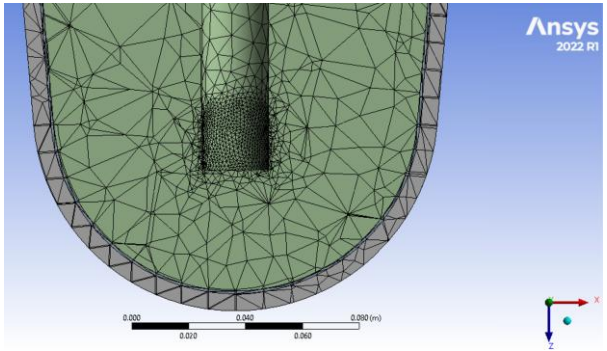


5 mm





2.5 mm

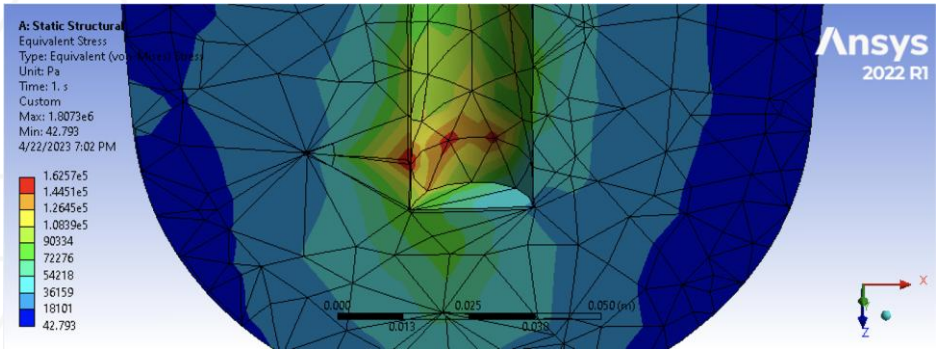


1.25 mm

- Sphere of influence around area of interest
- Decreased the mesh size by half until mesh convergence was reached

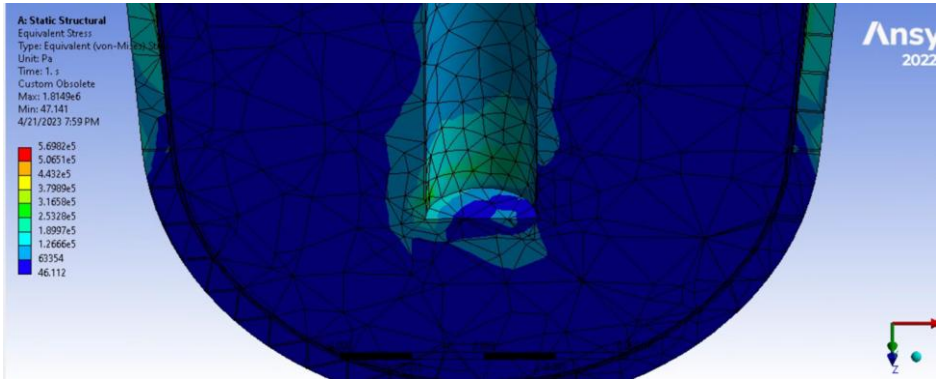
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MESH CONVERGENCE STUDY - RESULTS



10 mm

	Time [s]	Stress Probe Soft Tissue (Equivalent (von-Mises)) [Pa]
1	0.1	35523
2	0.2	48621
3	0.3	67081
4	0.4	82528
5	0.5	95942
6	0.6	1.1141e+005
7	0.7	1.2625e+005
8	0.8	1.4071e+005
9	0.9	1.5262e+005
10	1.	1.6519e+005

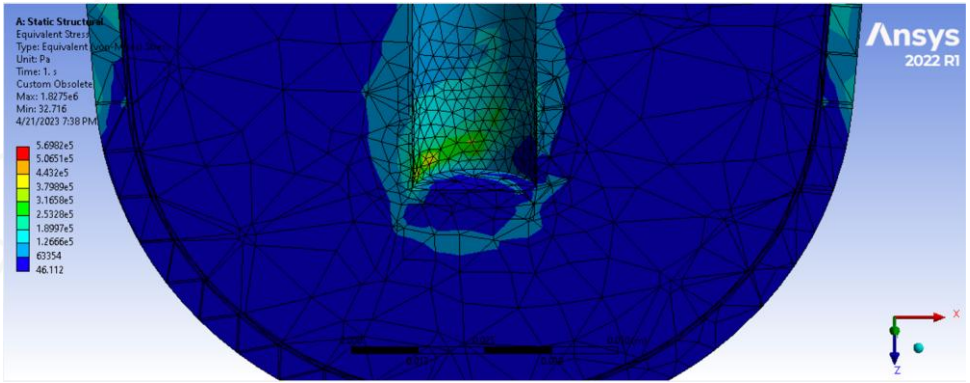


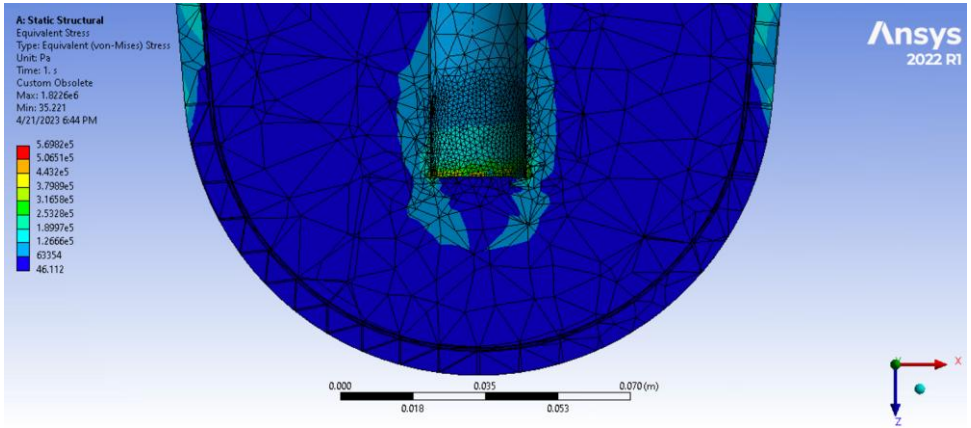
5 mm

	Time [s]	Stress Probe Soft Tissue (Equivalent (von-Mises)) [Pa]
1	0.1	32706
2	0.2	57211
3	0.3	1.136e+005
4	0.4	1.7731e+005
5	0.5	2.1761e+005
6	0.6	2.5407e+005
7	0.7	2.8761e+005
8	0.8	3.0708e+005
9	0.9	3.3433e+005
10	1.	3.5786e+005

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MESH CONVERGENCE STUDY - RESULTS





2.5 mm

	Time [s]	Stress Probe Soft Tissue (Equivalent (von-Mises)) [Pa]
1	0.1	36711
2	0.2	66607
3	0.3	1.4035e+005
4	0.4	2.0617e+005
5	0.5	2.6353e+005
6	0.6	3.2619e+005
7	0.7	3.8425e+005
8	0.8	4.3439e+005
9	0.9	4.7702e+005
10	1.	5.1778e+005

1.25 mm

	Time [s]	Stress Probe Soft Tissue (Equivalent (von-Mises)) [...
1	0.1	92854
2	0.2	1.631e+005
3	0.3	1.8567e+005
4	0.4	2.3672e+005
5	0.5	2.7179e+005
6	0.6	3.4586e+005
7	0.7	4.2022e+005
8	0.8	4.8459e+005
9	0.9	5.2528e+005
10	1.	5.6982e+005

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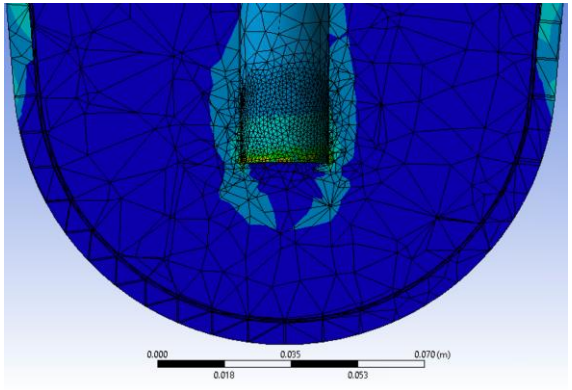
MESH CONVERGENCE - RESULTS

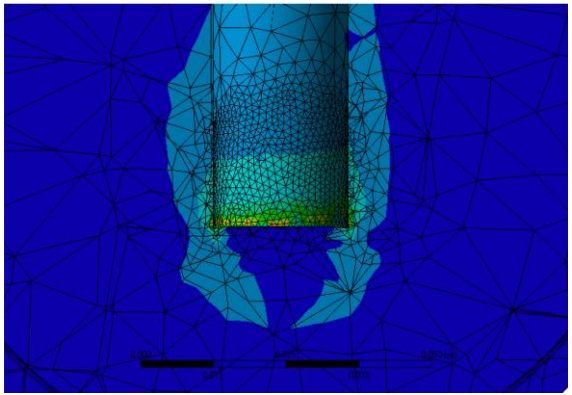
Mesh size (m)	Maximum equivalent von-Mises stress	RMS
0.01	1.65E+05 (Pa)	
0.005	3.58E+05	31.7%
0.0025	5.18E+05	16.8%
0.00125	5.698E+05	4.7%

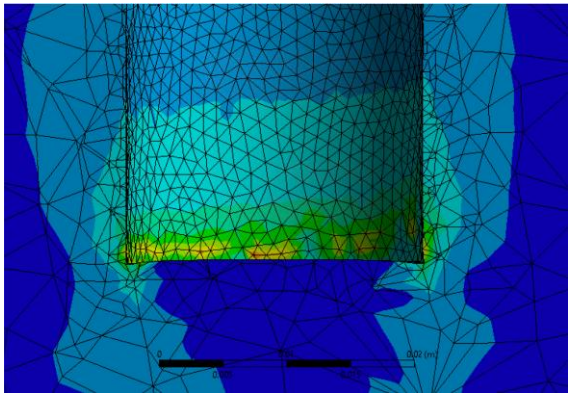
$$RMS = \frac{\frac{\sqrt{Old^2+New^2}-Old}{2}}{New}$$

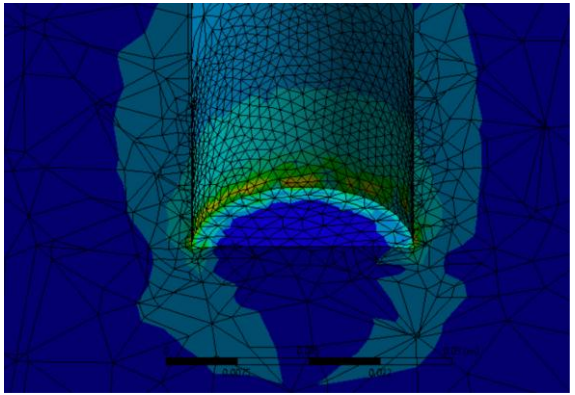
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RESULTS - CONVERGED MESH SIZE



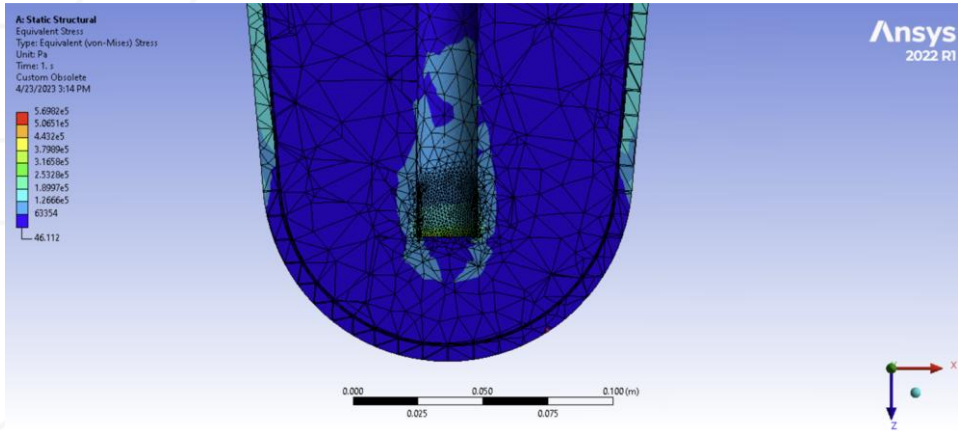


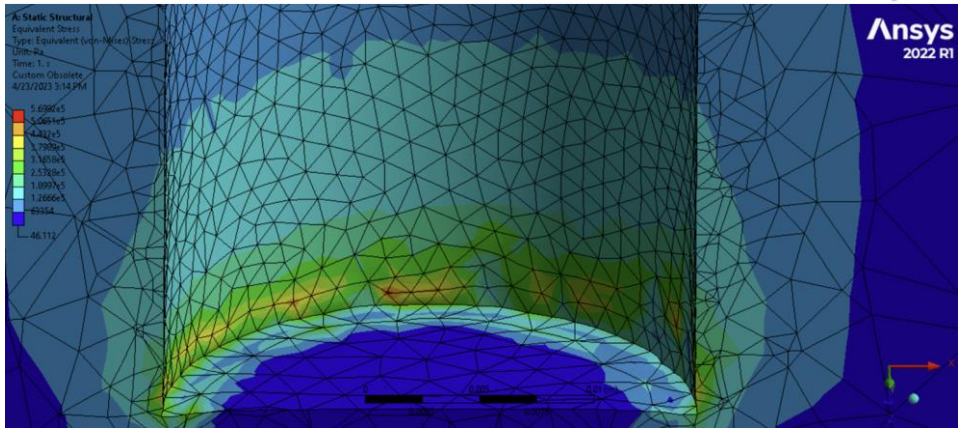




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SECOND STUDY - HYPERELASTIC MATERIAL



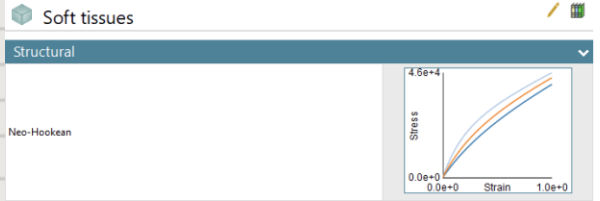


Hyper Elastic 1.25 mm

	Time [s]	Stress Probe st (Equivalent (von-Mises)) [Pa]
1	0.1	60600
2	0.2	1.2432e+005
3	0.3	1.4592e+005
4	0.4	2.2065e+005
5	0.5	2.7336e+005
6	0.6	3.2835e+005
7	0.7	3.9144e+005
8	0.8	4.411e+005
9	0.9	4.8935e+005
10	1.	5.409e+005

Linear Elastic 1.25 mm

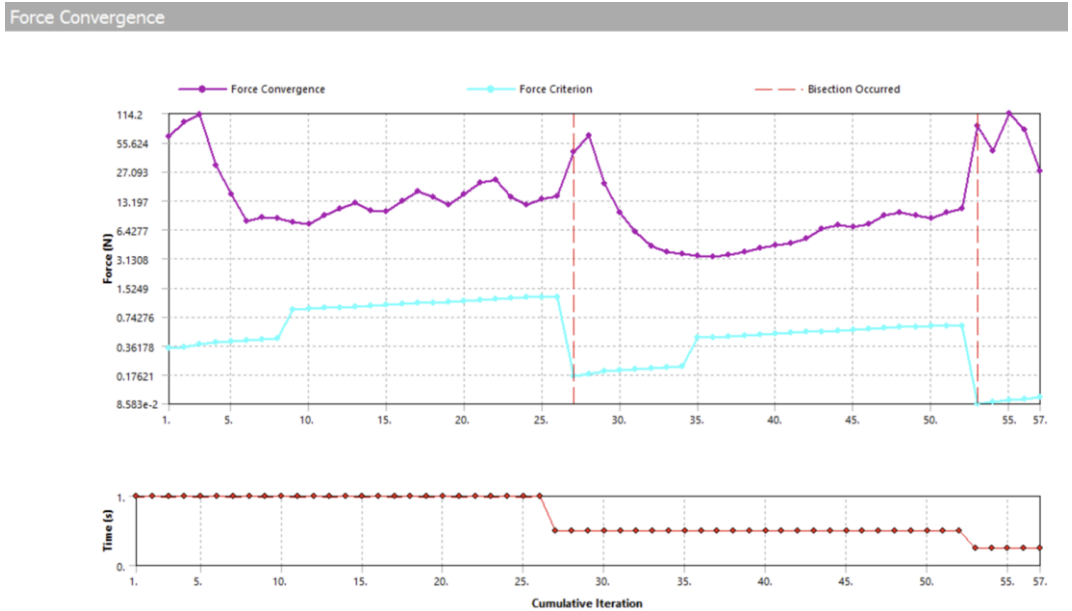
	Time [s]	Stress Probe Soft Tissue (Equivalent (von-Mises)) [...
1	0.1	92854
2	0.2	1.631e+005
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6	0.6	3.4586e+005
7	0.7	4.2022e+005
8	0.8	4.8459e+005
9	0.9	5.2528e+005
10	1.	5.6982e+005



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SECOND STUDY - MESH CONVERGENCE TRIAL

- Hyper Elastic Material: Bisection occurs consistently when using the Hyper Elastic Material model for soft tissue at higher mesh sizes than 1.25 mm



VALIDATION

- Met with Sujay Kestur, from the Exoskeleton and Prosthetic Intelligent Controls (EPIC) Lab at Georgia Tech
  - Unfortunately, Sujay only works on skeleton analysis and was unable to provide data for soft tissues analysis, but useful for force data for a future gait analysis

- So far, in the literature, the major method to validate the modelling is to compare the stresses predicted by the FE model with those measured experimentally

- Since we were unable to get experimental data from the EPIC lab, we compared our results with values found in the literature. We obtained a maximum stress in the limb of 540 kPa, while Zhang et al. had a value of 283 kPa. This different can be due to multiple factors:
  - We assumed the entire body weight being put in the limb when in reality, some is absorbed by the other leg while standing
  - Over-simplification of our geometry which decreases accuracy

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PROBLEMS ENCOUNTERED DURING OUR STUDY

- Too much penetration between contact
  - The normal stiffness factors of the contacts impacted were increased
  - Gave better accuracy and less penetration

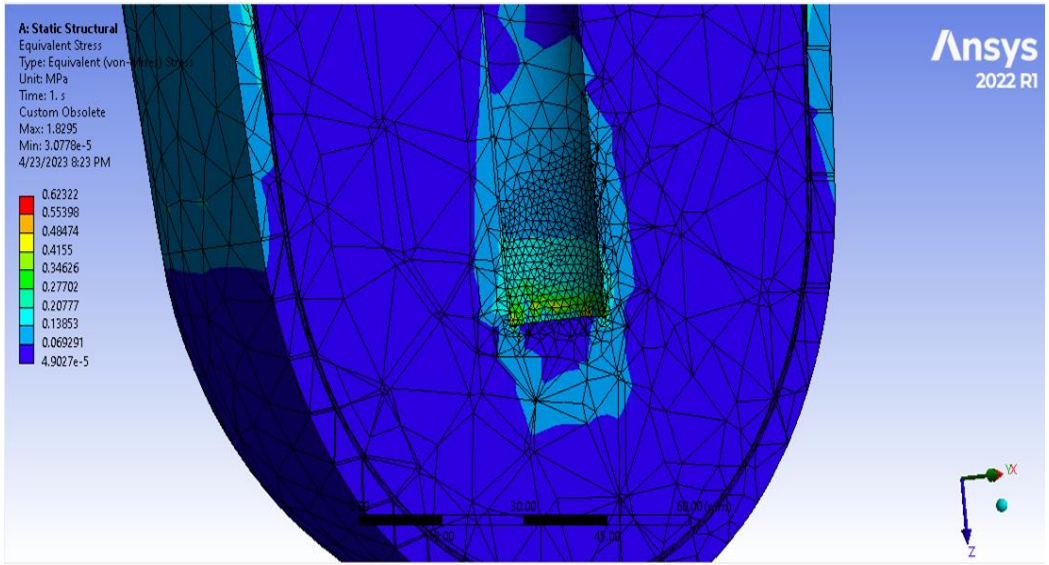
- Force convergence issue
  - Some contacts had too high of a stiffness factor, which led to the model oscillating and not converging
  - The normal stiffness factor of the contacts impacted were decreased

- Geometry Singularity
  - Bone has a 90 degree angle

- Because of a lot of convergence issues, the mesh convergence study had to be conducted as multiple single studies and not as a parameter study
  - Allowed troubleshooting and access to description of failure from ANSYS

ADDITIONAL “WHAT IF” STUDY

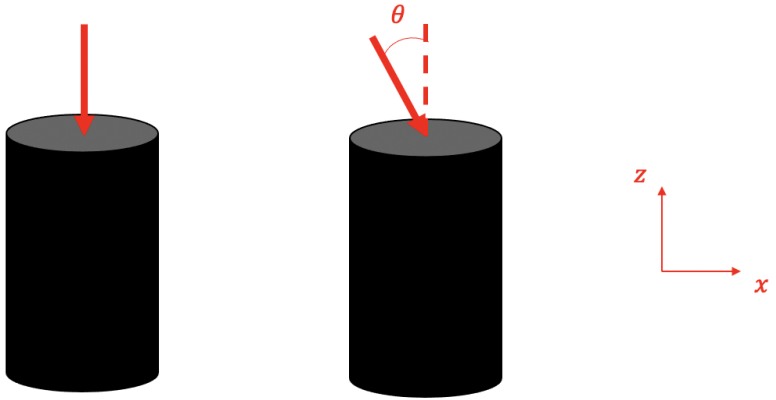
- Varied the material of the sock to evaluate its impact on the stresses in the limb
- Lowered Young’s Modulus but maintained Poisson’s ratio as .49
  - Decreased Y to .3 MPa and .25 MPa
  - Increased Stress measured at soft tissue probe



SUGGESTIONS FOR FUTURE RESEARCH

- Investigate the stresses in the residual limb during a 7-step gait analysis (simulates dynamic loading)
- Add additional layers between the residual limb and the socket
- Vary the thickness of the sock
- Conduct a thermal analysis to understand the thermic frictional effects near the limb/socket

7-Step Gait Analysis (Proposed Approach)



angles	fx gait	fz gait
-60	-679.66	-392.4
-45	-554.94	-554.94
-30	-392.4	-679.66
0	0	-784.8
30	392.4	-679.66
45	554.94	-554.94
60	679.66	-392.4

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CONCLUSION

- Model sufficient for understanding localization of stress within the residual limb of an upper leg amputee
- Hyperelastic modeling for soft tissue difficult when generating local contacts with multiple materials
- Further analysis of sock interaction required to improve model

Any Questions?

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Sources

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[3] Wu, J., Jin, Z., Jin, J., Yan, Y., & Tao, J. (2019). Study on the tensile modulus of seamless fabric and tight compression finite element modeling. Textile Research Journal, 90(1), 110–122. https://doi.org/10.1177/0040517519859931

[4] Portnoy S;Yarnitzky G;Yizhar Z;Kristal A;Oppenheim U;Siev-Ner I;Gefen A; (n.d.). Real-time patient-specific finite element analysis of internal stresses in the soft tissues of a residual limb: A new tool for prosthetic fitting. Annals of biomedical engineering. Retrieved April 23, 2023, from https://pubmed.ncbi.nlm.nih.gov/17120139/

[5] Affagard, J.-S., Feissel, P. & Bensamoun, S. F. Identification of hyperelastic properties of passive thigh muscle under compression with an inverse method from a displacement field measurement. J. Biomech. 48, 4081–4086 (2015).

[6] Author links open overlay panelJonathan Camargo a b, a, b, AbstractWe introduce a novel dataset containing 3-dimensional biomechanical and wearable sensor data from 22 able-bodied adults for multiple locomotion modes (level-ground/treadmill walking, Riener, R., Peng, J., Orendurff, M. S., Long, Y., Hu, B., Burden, A. M., Bohannon, R. W., Ankaralı, M. M., Camargo, J., Chvatal, S. A., Delp, S. L., Farris, D. J., Fukuchi, C. A., Fukuchi, R. K., Huang, H., & Hug, F. (2021, February 20). A comprehensive, open-source dataset of lower limb biomechanics in multiple conditions of stairs, ramps, and level-ground ambulation and Transitions. Journal of Biomechanics. Retrieved April 23, 2023, from https://www.sciencedirect.com/science/article/pii/S0021929021001007?via%3Dihub#s0055

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